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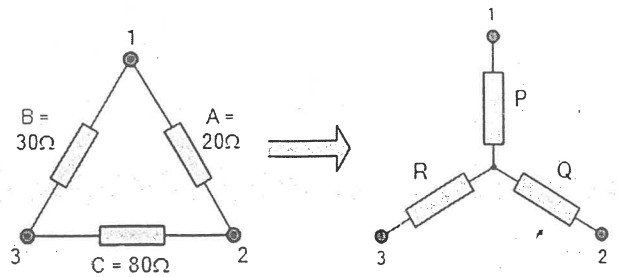
[23ESEE11]

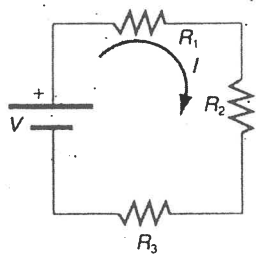
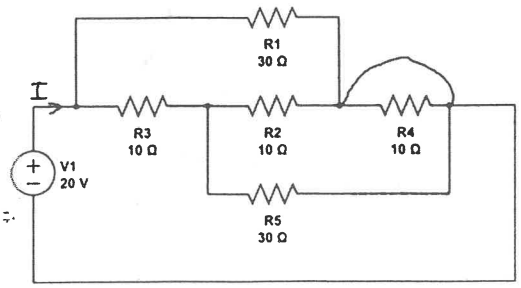
FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	05 X 02=10		
1.	**a**	State ohm's law and mention its equation	L1	Co1	02
	b.	State the right-hand thumb rule and cork screw rule	L2	Co2	02
	c.	Convert the following current from polar form to rectangular form $56\angle 27^{0}$ A	L2	Co3	02
	d.	Compare capacitors with batteries	L1	Co5	02
	e.	List the types of protection available for residential installation.	L1	Co6	02
		PART-B	06 X 05=30		
2.	**a.**	The resistance of a wire is 5Ω at 50°C and 6Ω at 100°C. Then what will be the resistance of the wire at 0°C?	L3	Co1	05
		OR			
	b.	Discuss on the charging and discharging of inductors and capacitors with neat illustrations. Give the voltage and current relations.	L2	Co1	05
3.	**a.**	State the laws related to electromagnetic induction and compare statically induced EMF with dynamically induced EMF.	L1	Co2	05
		OR			
	b.	Justify the need for self-inductance and mutual inductance hence give the coefficient of coupling of mutually connected coils.	L1	Co2	05
4.	**a.**	Draw a symmetrical sinusoidal voltage waveform with a magnitude and frequency of your choice and mark time period, peak voltage, peak to peak voltage, instantaneous voltage. What will be its average voltage?	L3	Co3	05
		OR			
	b.	Explain RL series circuit when it is connected to an ac source with a phasor diagram	L3	Co3	05
5.	**a.**	Convert the following resistances connected in Star to Delta. 	L3	Co4	05
		OR			

	b.	Mention the advantages of 3 phase systems over single phase systems	L1	Co4	05
6.	a.	Discuss in detail the basic principle of cellular communication	L2	Co5	05
		OR			
	b.	Illustrate the block diagram of a GSM communication system	L2	Co5	05
7.	a.	What is electric shock? How to give first aid to a person undergone an electric shock. List the safety rules to be followed while working with electrical installations.	L2	Co6	05
		OR			
	b.	Draw the diagram of grounding of electrical installation. List the materials used and their role.	L2	Co6	05
		PART-C	**06 X 10=60**		
8.	a.	The voltage output of the battery in Figure is 12.0 V, and the resistances are R1 = 1.00 Ω, R2 = 6.00 Ω, and R3 = 13.0 Ω. (a) What is the total resistance? (b) Find the current. (c) Calculate the voltage drop in each resistor, and show these add to equal the voltage output of the source. (d) Calculate the power dissipated by each resistor. (e) Find the power output of the source, and show that it equals the total power dissipated by the resistors. 	L3	Co1	10
		OR			
	b.	Calculate the current (I) flowing through the circuit as shown below, 	L3	Co1	10
9.	a.	Discuss the Statically Induced EMF and Dynamically Induced EMF with neat equations.	L2	CO2	10
		OR			
	b.	Justify the need of cross and dot conventions and explain Thumb rule, cork screw rule.	L2	Co2	10

10.	**a.**	Explain the concept of active, reactive, apparent power, average power and power factor.	L2	Co3	**10**
		OR			
	b.	Draw an AC signal and define its RMS and Average value. Also derive the expression for RMS value of alternating voltage	L3	Co3	**10**
11.	a.	A balanced star connected load of (8+6j) Ω per phase is connected to a 3Φ 230V supply. Determine the line current, power factor, power, reactive volt-ampere and total VA.	L3	CO4	**10**
		OR			
	b.	Illustrate the relationship between the line and phase values of voltage and current in a three phase, star connected circuit with relevant phasor diagram and circuit diagram.	L3	CO4	**10**
12.	**a.**	Draw the symbol, write the general specifications and types of capacitors and inductors	L2	Co5	**10**
		OR			
	b.	What is the name of the single digital equipment used to measure voltage, current and resistance? Give its basic components and explain their role.	L2	C05	**10**
13.	**a.**	Discuss on the various types of Residual Current breakers and give their advantages and drawbacks.	L2	Co6	**10**
		OR			
	b.	Compare and contrast on various types of electric lamps used in domestic application.	L2	Co6	**10**